ST117 Individual DRAFT Written Report - Phase 2 Task 3 (Woodland vegetation - Butterflies and Moths)

Report Pod MS-10

2026-05-09

a)

```r
# Initialising
MothsDF <- read.csv("Invertebrates/ECN_IM1.csv")
ButterflyDF <- read.csv("Invertebrates/ECN_IB1.csv")
WoodlandDF <- read.csv("Vegetation/ECN_VW1.csv")

# First we'll add a year column
MothsDF$SYEAR <- year(dmy(MothsDF$SDATE))
ButterflyDF$SMONTH <- month(dmy(ButterflyDF$SDATE))
ButterflyDF$SYEAR <- year(dmy(ButterflyDF$SDATE))
# WoodlandDF already has SYEAR


# Now removing 0 values and NAs
MothsDF <- filter(MothsDF, VALUE >0 & !is.na(VALUE))
ButterflyDF <- filter(ButterflyDF, VALUE > 0 & !is.na(VALUE))
WoodlandDF <- filter(WoodlandDF, !is.na(TREE_SPEC))

# Filtering out March and October entries as these shouldn't have been recorded
# See phase 1
ButterflyDF <- filter(ButterflyDF, SMONTH >= 4 & SMONTH <= 9)


# Aggregating data to get totals for each year and site
butterflyTotal <- aggregate(VALUE ~ SITECODE + SYEAR, data=ButterflyDF, FUN=sum)
mothTotal <- aggregate(VALUE ~ SITECODE + SYEAR, data=MothsDF, FUN=sum)
woodlandTotal <- aggregate(TREE_SPEC ~ SITECODE + SYEAR, data=WoodlandDF, FUN=function(x) length(uni



# Left joining the butterfly onto moth one
invCombined <- butterflyTotal |> left_join(mothTotal, by=c("SITECODE", "SYEAR"))

# Now adding woodland data to the above
part3aDF <- invCombined |> left_join(woodlandTotal, by=c("SITECODE", "SYEAR"))


# Renaming columns for readability
colnames(part3aDF) <- c("SITECODE", "SYEAR", "ButterflyTotal", "MothTotal", "WoodlandTotalSpecies")

head(part3aDF, 3)
```

```
##   SITECODE SYEAR ButterflyTotal MothTotal WoodlandTotalSpecies
## 1      T01  1993            302      3141                   NA
## 2      T03  1993             43       581                   NA
## 3      T04  1993             91       454                   NA
```

```
# Repeat for moths
part3aMothDF <- mothTotal |> left_join(woodlandTotal, by=c("SITECODE", "SYEAR"))
colnames(part3aMothDF) <- c("SITECODE", "SYEAR", "MothTotal", "WoodlandTotalSpecies")

head(part3aMothDF, 3)
```

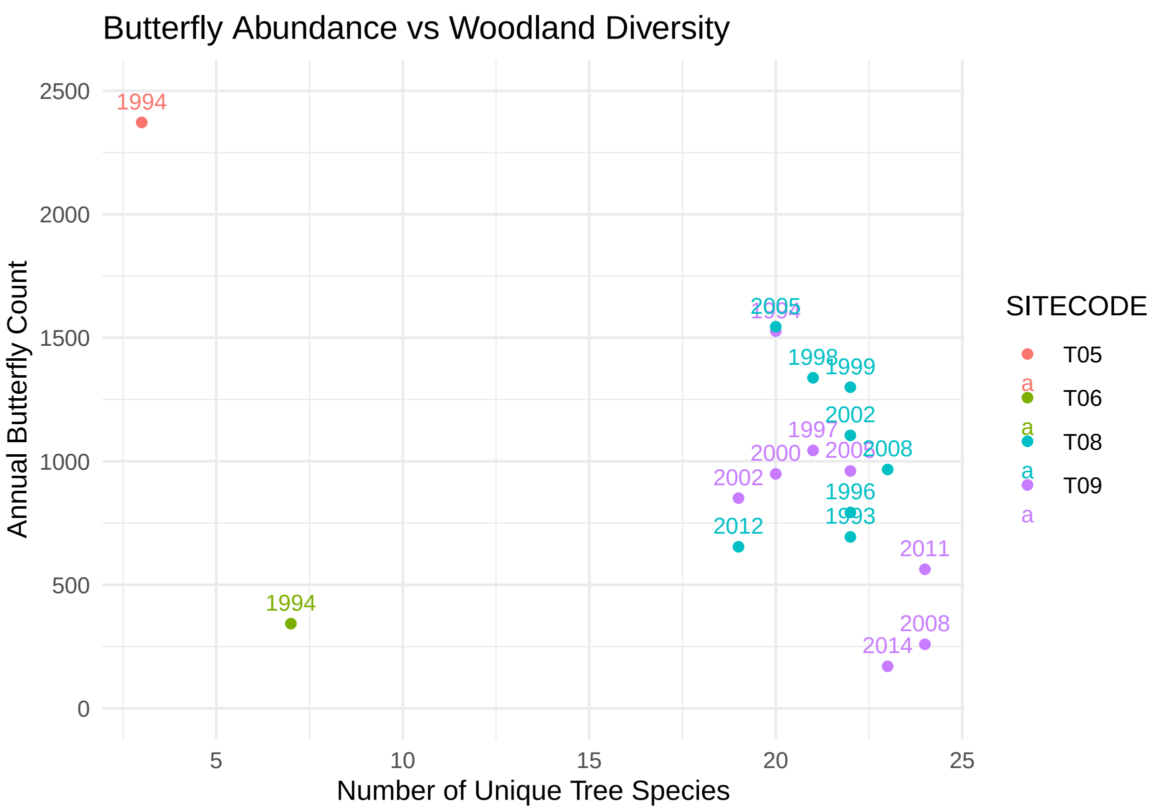
```
##   SITECODE SYEAR MothTotal WoodlandTotalSpecies
## 1      T06  1992      1788                   NA
## 2      T08  1992        13                   NA
## 3      T09  1992       303                   NA
```

I didn't expect there to be so many NA values under WoodlandTotalSpecies. Out of 227 annual records for the insects, only 18 of these match with the available woodland data. I suppose this makes sense given how the woodland data was attained. The same pattern holds for moths.

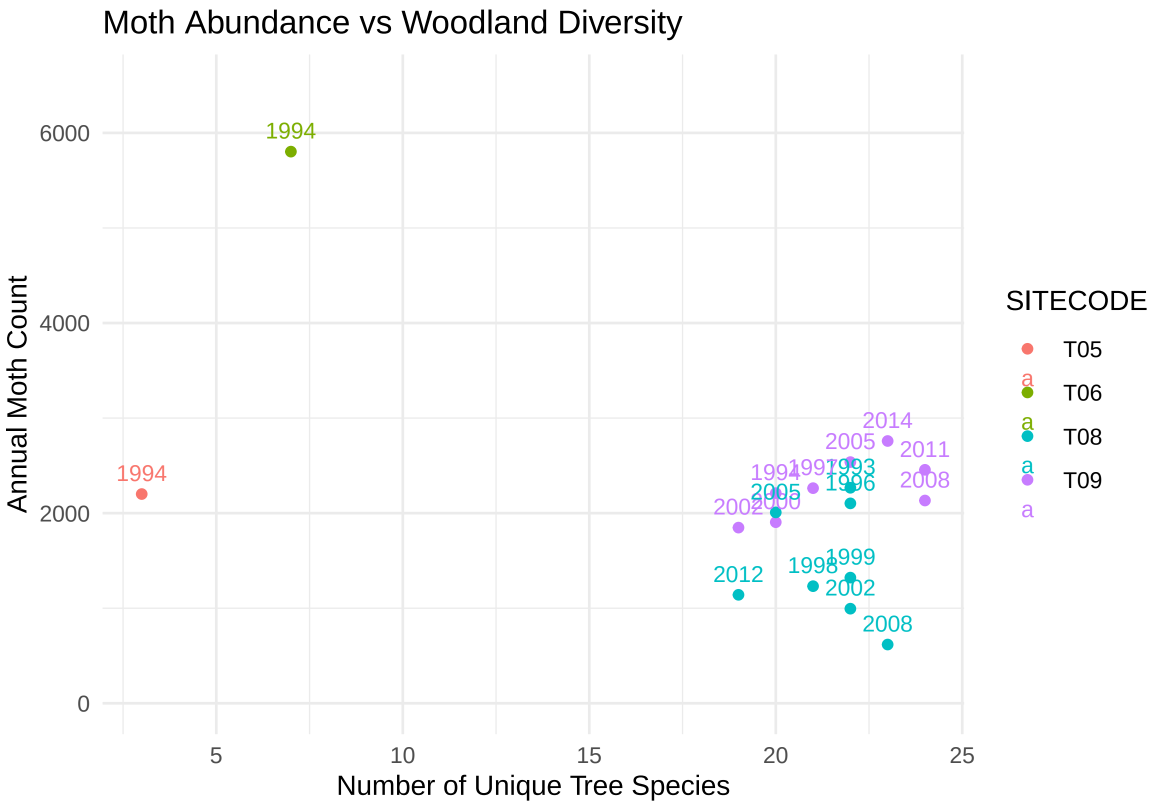
b)

```
# For the plot, I don't need to bother with the na values
part3bPlotData <- filter(part3aDF, !is.na(WoodlandTotalSpecies))
part3bPlotData$SITECODE <- as.factor(part3bPlotData$SITECODE)

ggplot(part3bPlotData, aes(x=WoodlandTotalSpecies, y=ButterflyTotal, color=SITECODE)) +
  geom_point() + geom_text(aes(label=SYEAR), vjust=-0.8, size=3) +     theme_minimal() +
  labs(title="Butterfly Abundance vs Woodland Diversity",
       x="Number of Unique Tree Species",
       y="Annual Butterfly Count") + ylim(0,2500)
```




```
ggplot(part3bPlotData, aes(x=WoodlandTotalSpecies, y=MothTotal, color=SITECODE)) +
  geom_point() + theme_minimal() +
  geom_text(aes(label=SYEAR), vjust=-0.8, size=3) +
  labs(title="Moth Abundance vs Woodland Diversity",
       x="Number of Unique Tree Species",
       y="Annual Moth Count") + ylim(0,6500)
```



Neither scatterplot shows a clear linear relationship between woodland species diversity and invertebrate counts. The data points cluster by site, suggesting the differing sites are causing most of the variation. Sites 5 and 6 in 1994 stand out as outliers in both plots, sitting far from the main cluster.

c)

The first potential issue for fitting a linear regression model is the very small sample size. Although there were 227 annual insect records originally, only 18 remained after matching with woodland vegetation data. This makes it difficult to draw reliable conclusions.

Secondly, the scatterplots show some clustering dependent on sites. Different sites appear to follow different patterns, violating the assumption that a single linear relationship explains all observations. There are also some clear outliers, particularly sites 5 and 6 in 1994, which would heavily influence the fitted regression line.

d)

```r
# Built in linear regression function
model_inv <- lm(MothTotal ~ ButterflyTotal, data=part3aDF)

# Summary table for pooled model
kable(data.frame(Term = c("Intercept", "ButterflyTotal"),
  Estimate = round(summary(model_inv)$coefficients[, 1], 3),
  Std.Error = round(summary(model_inv)$coefficients[, 2], 3),
  p.value = round(summary(model_inv)$coefficients[, 4], 3)), caption = "Pooled Model Coefficients: M
```

Table 1: Pooled Model Coefficients: MothTotal ~ ButterflyTotal

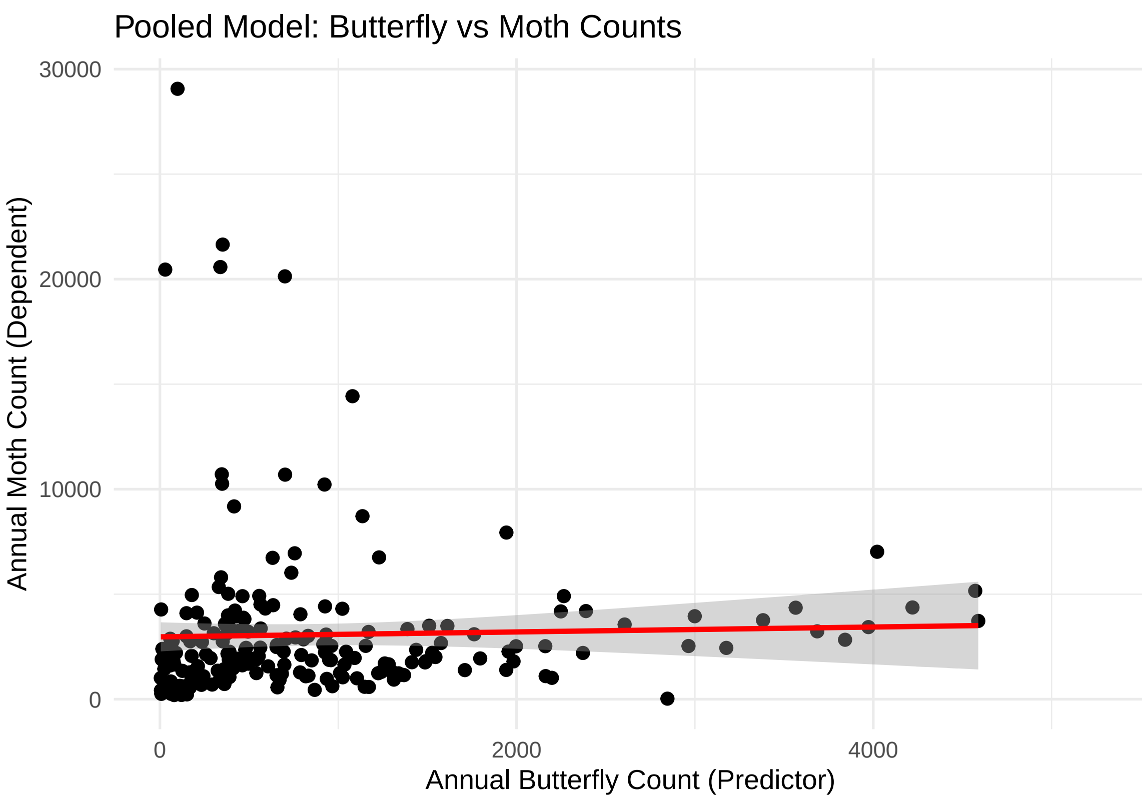
Term	Estimate	Std.Error	p.value
Intercept	2968.765	351.873	0.000
ButterflyTotal	0.117	0.276	0.672

```r
ggplot(part3aDF, aes(x = ButterflyTotal, y = MothTotal)) +
  geom_point(size=2) +
  geom_smooth(method="lm", color="red") +
  theme_minimal() +
  labs(title = "Pooled Model: Butterfly vs Moth Counts",
       x = "Annual Butterfly Count (Predictor)",
       y = "Annual Moth Count (Dependent)")
```

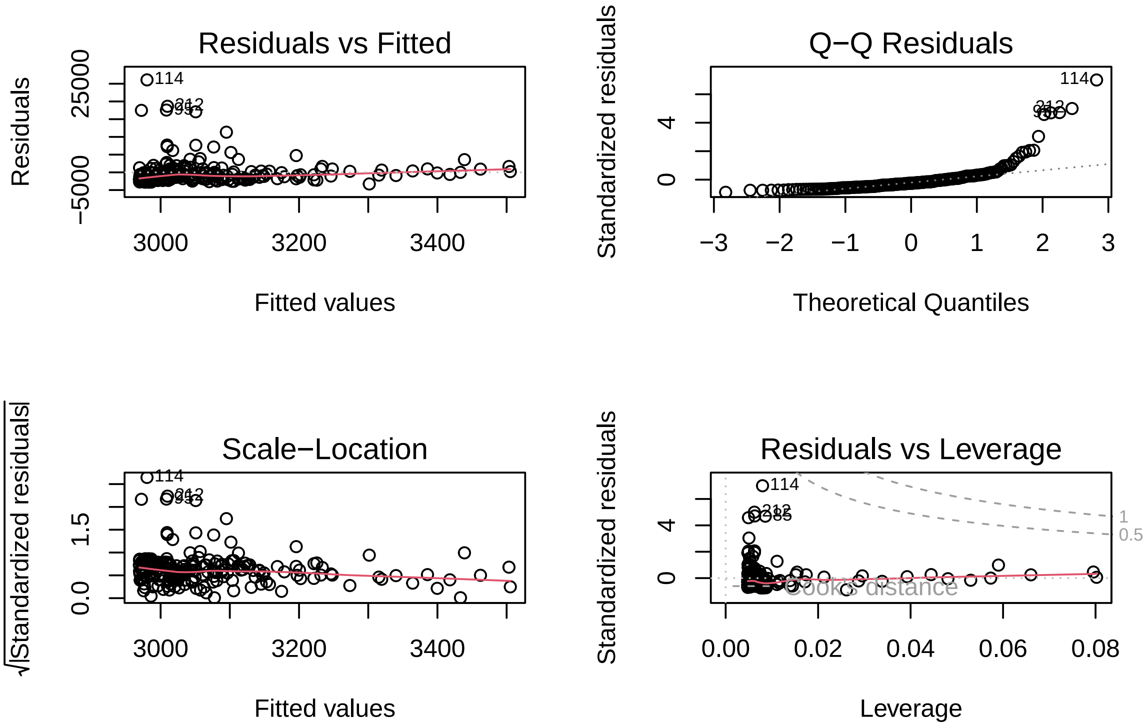
```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 18 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 18 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
par(mfrow=c(2,2))
plot(model_inv)
```




```r
# Remove point 114 and refit
part3aDF_clean <- part3aDF[-114, ]
model_inv_clean <- lm(MothTotal ~ ButterflyTotal, data=part3aDF_clean)

kable(data.frame(
  Term = c("Intercept", "ButterflyTotal"),
  Estimate = round(coef(model_inv_clean), 3),
  Std.Error = round(summary(model_inv_clean)$coefficients[,2], 3),
  p.value = round(summary(model_inv_clean)$coefficients[,4], 3)
), caption = "Refitted Model Coefficients (outlier removed): MothTotal ~ ButterflyTotal", row.names=
```

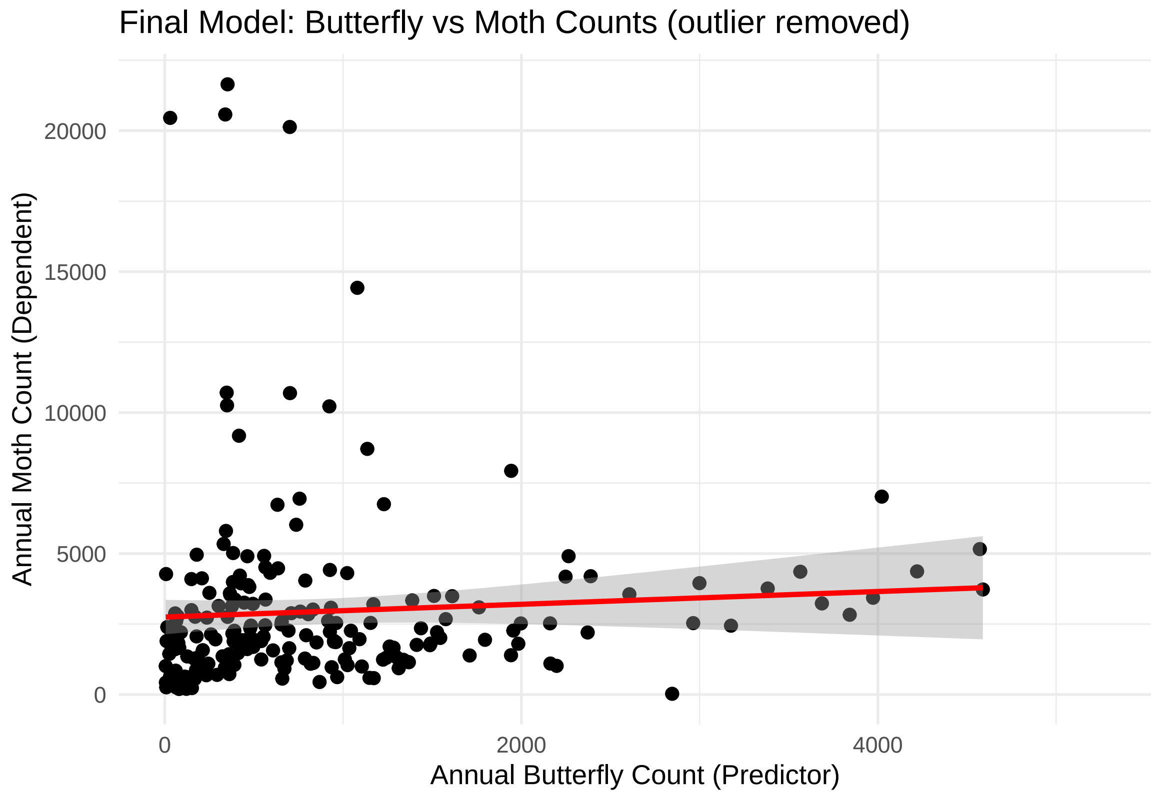
Table 2: Refitted Model Coefficients (outlier removed): MothTotal ~ ButterflyTotal

Term	Estimate	Std.Error	p.value
Intercept	2748.282	309.368	0.00
ButterflyTotal	0.226	0.242	0.35

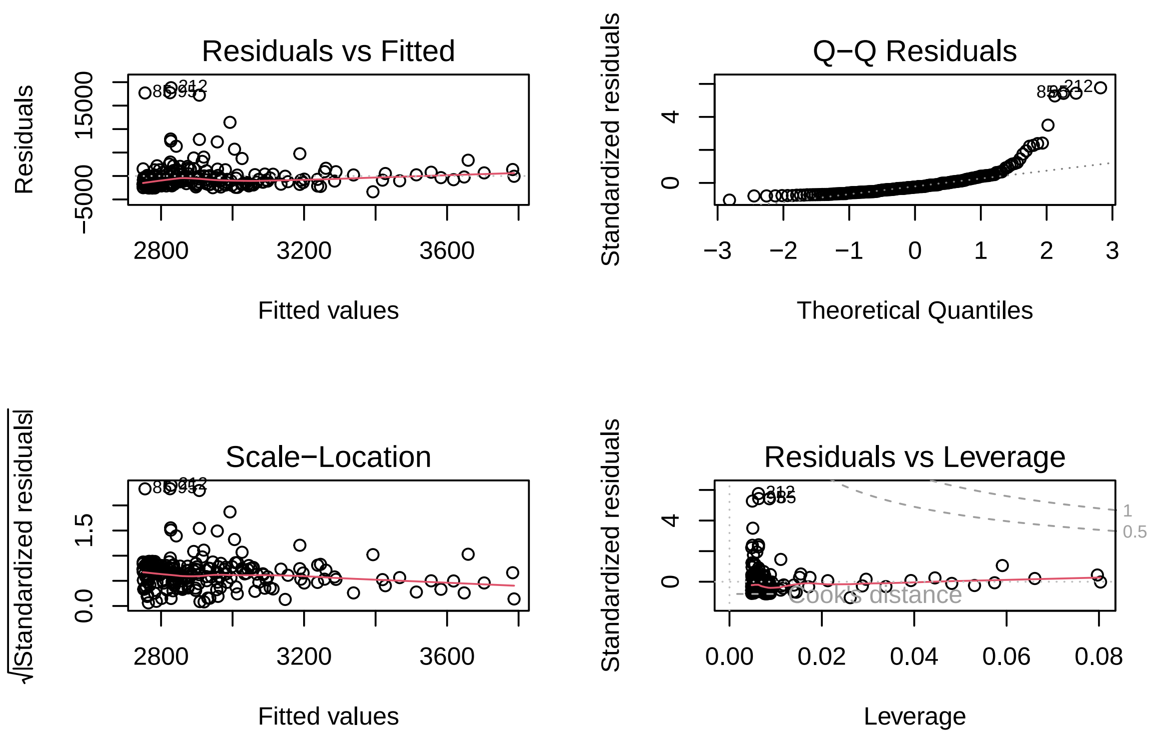
```r
ggplot(part3aDF_clean, aes(x = ButterflyTotal, y = MothTotal)) +
  geom_point(size=2) +
  geom_smooth(method="lm", color="red") +
  theme_minimal() +
  labs(title = "Final Model: Butterfly vs Moth Counts (outlier removed)",
       x = "Annual Butterfly Count (Predictor)",
       y = "Annual Moth Count (Dependent)")
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 18 rows containing non-finite outside the scale range
## (`stat_smooth()`).
## Removed 18 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
par(mfrow=c(2,2))
plot(model_inv_clean)
```



For the model fitting process, I used the built in function lm() to fit a linear regression model with MothTotal as the dependent variable and ButterflyTotal as the predictor. The parameters for the creating the line were estimated using the method of least squares, which seeks to minimise the distance from points to line. The R^2 value is very low at 0.0008673, suggesting that the model explains almost none of the variance in moth counts. The p-value is also high at 0.6721, indicating that butterfly counts are not a significant predictor of moths in this dataset.

I then check the residuals (the difference between the observed and fitted values). First, observe the residuals vs fitted plot. The red line here is fairly straight, which may slightly support the assumption of a linear relationship.
Next, observe the Q-Q plot. There is some extreme deviation from the line at the right tail.

Interestingly, the same points which are outliers in this plot are also outliers in the other plots. This could indicate that the residuals are not normally distributed, thanks to some extreme outliers in the moth population. Next, observe the Scale-Location plot. Our lecture notes claim that residuals should ideally be spread equally along the range. Clearly, we see a much larger spread on the left, suggesting that the variance is not constant.
Finally, observe the Residuals vs Leverage plot. This lays bare the culprits which have overly impacted the regression line. In particular, see point 114, which appears to be the biggest outlier across all plots.

Perhaps one of the most immediate improvements could be removing outliers, such as the aforementioned point 114. It has high leverage, meaning that it has had a great impact on the final line. Or, looking back at the plot in task 3b, we do see that certain sites tend to cluster together significantly. Perhaps we could separate the data points according to their sites to see if different sites follow different regression lines.

I refit the model after removing point 114. The R^2 remains very low and the relationship remains insignificant, confirming that the poor fit is not just down to this one outlier. In conclusion, butterfly counts do not appear to be a useful predictor of moth counts in this dataset.